Twelfth Annual ECC Undergraduate Mathematics Competition, April 4, 2009

No CALCULATORS, COMPUTERS, BOOKS, NOTES or NON-TEAM-MEMBERS may be consulted.

PLEASE BEGIN EACH PROBLEM ON A NEW SHEET OF PAPER. Team identification and problem number should be clearly given at the top of each sheet of paper submitted.

Each problem counts 10 points. Partial credit will be given for incomplete but significant work. For full credit, answers must be fully justified. (Which in some cases may simply mean showing all work and reasoning.) Have fun!

* * Time control: three hours * *

1. Solve for x.

Find all real numbers x satisfying

$$\log((4^3x^25^3)^{-3}) = -9,$$

where "log" denotes the common (base 10) logarithm.

2. A multiple of 5.

Prove that $1^{2009} + 2^{2009} + 3^{2009} + 4^{2009} + 5^{2009}$ is an integral multiple of 5.

3. Minimum value.

Find the minimum value of the function f: $\boldsymbol{R}\to\boldsymbol{R}$ defined by

$$f(y) = \int_0^1 [x^2 - (x+y)]^2 dx.$$

4. Tower of Babel.

Suppose that in year 1 humankind began rebuilding the Tower of Babel from the ground up, and each year extended its height by the number of meters equal to the product of the digits of the year number. Thus, in year 1 its height grows by 1 meter; in year 1492, by $1 \cdot 4 \cdot 9 \cdot 2 = 72$ meters. How high would the tower be at the end of year 2009?

5. Sum of squares of the roots.

Let r, s and t be the roots of the equation $x^3 + 3x^2 - 5x - 1 = 0$. Find $r^2 + s^2 + t^2$.

6. Urns and balls.

An urn contains 3 red balls and 5 white balls. Balls are drawn at random, one at a time, from the urn, and not replaced. Find the probability that it takes more than four draws to remove all three red balls.

7. A double sequence.

The sequence (x_n, y_n) of real number pairs is defined by

$$(x_0, y_0) = (1, 0) \quad \text{and for } n \geq 0,\ (x_{n+1}, y_{n+1}) = (x_n + 2y_n, x_n + y_n).$$

Find, with proof, the value of $x_{2009}^2 - 2y_{2009}^2$.

8. Area of a triangle.

The altitudes of a triangle are 20, 24 and 30. Find the area of the triangle.

9. Convergence of a series.

Let $z_1, z_2, z_3, \ldots$ be a sequence of complex numbers all having nonnegative real part. Given that the series

$$\sum_{k=1}^{\infty} z_k \quad \text{and} \quad \sum_{k=1}^{\infty} z_k^2$$

both converge, prove that

$$\sum_{k=1}^{\infty} |z_k|^2$$

converges.

10. Solutions in integers.

Find all pairs (m, n) of positive integers satisfying

$$3 \cdot 2^m + 1 = n^2$$

and show that there are no more.